

Lecture 1: Introduction to Algorithms and their Limitations

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What is an Algorithm?

- Steps to solve a problem
 - Procedure given inputs $\rightarrow$ outputs
 - Simplification of a “real-world way to solve a problem”
 - Methodology to manage a data set
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Course Structure

Unit	Topic
1	Storing & Searching
2	Computational Models
3	Graph & Logic
4/5	Computational Complexity & Uncomputability

Unit 1 — Goals

- Abstracting a computational problem
 - Prove an algorithm solves a problem
 - Analyze and compare efficiency
 - Formalize “what is an algorithm”
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Motivating Example: Keyword Search on Google

PageRank: $\text{PR}(\text{url}) \in [0, 1]$ for every url on the WWW

Keyword search: Given word w , consider $S_w = \{\text{all urls where } w \text{ is found}\}$

Output: Sorted list — return urls in S_w in decreasing order of $\text{PR}(\cdot)$

The Sorting Problem

Input: $A = \{(K_0, V_0), (K_1, V_1), \dots, (K_{n-1}, V_{n-1})\}$ where each $K_i \in \mathbb{R}$

Output: Array $A' = \{(K'_0, V'_0), (K'_1, V'_1), \dots, (K'_{n-1}, V'_{n-1})\}$ such that A' is a *valid sort* of A :

- $K'_0 \leq K'_1 \leq \dots \leq K'_{n-1}$
 - $\exists$ permutation $\pi : [n] \rightarrow [n]$ such that $(K'_i, V'_i) = (K_{\pi(i)}, V_{\pi(i)})$
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Applying the Sorting Abstraction to Google Search

From S_w , define:

$$K_i = 1 - \text{PR}(\text{url}_i), \quad V_i = \text{url}_i$$

Sorting by K_i gives urls in decreasing order of PageRank.

Insertion Sort

InsertionSort(A):

For all $i = 0$ to $n - 1$:

1. Find the smallest $j \leq i$ such that $A[j][0] \geq A[i][0]$
 2. Insert $A[i]$ at position j and shift $A[j \dots i - 1]$ to $A[j + 1 \dots i]$
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Insertion Sort — Example

Input: $(6, a), (2, b), (1, c), (9, d)$

i	Array state
0	$(6, a), (2, b), (1, c), (9, d)$
1	$(2, b), (6, a), (1, c), (9, d)$
2	$(1, c), (2, b), (6, a), (9, d)$
3	$(1, c), (2, b), (6, a), (9, d)$
